

UGANTHAN M

✉ uganthanmariappan@gmail.com ☎ +91 8667670398 📄 uganthanm 🕒 uganthan2005 🏠 uganthanmariappan

Education

B.Tech ARTIFICIAL INTELLIGENCE AND DATA SCIENCE, CHENNAI INSTITUTE OF TECHNOLOGY	2023 – present
CGPA : 8.8/10	CHENNAI, INDIA
B.Sc DATA SCIENCE (FOUNDATION), INDIAN INSTITUTE OF TECHNOLOGY, MADRAS	2024/09 – present
CGPA : 7.4/10	CHENNAI, INDIA

Professional Experience

INTERN - Crayon'd GenAI Skill Showcase Program (May-June 2025) 🔗

- Built a **Movie Context Analyzer** using Gemini, DeepSeek & Qwen; achieved **~25% higher accuracy** via advanced prompt optimization.
- Built a **RAG-based Document QA Bot** with chunking, embeddings, and Supabase vector search, improving **retrieval precision by 35%**.
- Developed a **LangGraph - Report Maker & Analyzer** with interactive feedback loops, enhancing **user validation efficiency by 40%**.
- Engineered a **Voice-Based Multi-Agent System** integrating STT/TTS and modular AI agents, boosting **real-time interaction by 30%**.

Projects

NodeCode IDE 🔗, Next.js 15+, TypeScript, React Flow, Node.js, Prisma ORM, PostgreSQL, Docker, xterm.js

- A **bi-directional** visual development environment that maps **TypeScript codebases into interactive node graphs** for **real-time editing** and **AI-powered** repository scaffolding.
- Built a **Single Source of Truth (SSoT)** architecture using a JSON schema and **ts-morph** for **deterministic, corruption-free round-trip code synchronization**. Added a split-screen visual editor, WebSocket terminal, and a **two-stage AI pipeline** for architecture-first code generation.
- **70% faster onboarding**: Reduced legacy codebase onboarding from weeks to hours with visual code mapping.

EchoLink - Private Multi Device Mesh Network 🔗, Avalonia, C#, .NET 10, SSH.NET, Tailscale, Headscale, Go (Android bridge)

- Built a **C#/.NET 10 Macro engine** supporting dual-mode (Terminal/GUI) execution on **Windows and Linux**, cutting administrative latency by **40%**.
- **Architected real-time monitoring and camera sharing** features using asynchronous data streams to maintain **<100ms latency** via Wireguard.
- **Co-developed a zero-trust P2P system** using Tailscale and a Go Android Bridge for **100% cloud-independent** file and audio streaming.

EEG Emotion Analysis 🔗, Python, Numpy, Pandas, Matplotlib, MNE python, Seaborn, TensorFlow / Keras, scikit-learn

- Developed a **Deep Learning Model** for **EEG-based emotion detection** using **LSTM networks** to capture temporal signal patterns.
- Implemented **data preprocessing and feature extraction pipelines** to enhance signal clarity and model performance.
- Benchmarked against **KNN, Linear Regression, and other ML models**, validating performance through statistical evaluation.
- Achieved **~22% higher classification accuracy** with LSTM, demonstrating the **superiority of deep sequential modeling** for neurodata interpretation.

RAG-based Q&A system, Python, PyMuPDF Google Gemini API, Supabase (pgvector), NumPy

- Built an **End-to-End RAG Pipeline** to **extract, chunk, and embed PDFs** using **Gemini embeddings** for semantic retrieval.
- Stored **vector embeddings in Supabase (pgvector)** to enable **high-speed retrieval and contextually relevant search**.
- Implemented a **retrieval-augmented Q&A model** powered by **Gemini LLM** for context-grounded response generation.
- Achieved **~35% improvement in response precision and factual alignment**, demonstrating expertise in **knowledge-grounded AI workflow design**.

Skills

Programming: Python, C++, pgSQL, JavaScript, C#, Go, Kotlin, Dart.
Web Development: HTML, CSS, Tailwind CSS, React, Node JS, FAST API.
Database: PostgreSQL, SQL, MongoDB, Supabase, ChromaDB.
Designing: Figma, Canva, Behance.

Machine Learning/Data Analysis: Python- SciKit Learn, Seaborn, Pandas, NumPy, Matplotlib, Tableau, Excel, PyTorch, Tensorflow.
Developer Tools: Visual Studio Code, Chrome Developer Tools, Jupyter, Google Colab, Git, Github, Docker.

Volunteering

Technical Head, TALOS 5.0 - 2026 🔗

- **Architected a full-stack registration portal (Next.js, Python, Firebase)** that processed payments and data for over **2,000+ teams**. Led the operational **execution and resource allocation** for **10 competitive events and 5 workshops**. **Delivered** a specialized **workshop on MCP and AI**, training participants on emerging concepts in **agentic workflows and automation**.

Graphic Designer - Asymmetric - [2024- Present]

- **Designed** end-to-end marketing assets including **posters, merchandise (T-shirts), and promotional videos** for flagship events like **Tech Fiesta and Hacksymmetric** and **Mentored** and led a team of **junior designers** to ensure high-**quality visual** output and brand consistency across all club events.

Achievements and Coding Proficiency

- **Shortlisted** for **Smart India Hackathon (SIH) Internal Hackathon** - **Top 25 teams** selected within the college.
- **TCS CodeVita Season 13:** Secured a **Global Rank of 1304** in **Round 1**, qualifying among the top percentile of participants in one of the world's largest coding contests.
- **LEETCODE:(uganthan_m)** Top **5.48% Globally**; Solved **450+** sums; **Max Ratings: 1,861 (Knight Badge)**. 🔗
- **CODECHEF:(uganthan_m)** **3 star**; Max Rating **1635**; Best Contest **Rank 169** 🔗 .

Certifications

Android developers intern certificate - Eduskill 🔗 | Parallel Computer Architecture - NPTEL 🔗 | CCNA (Introduction to Networks) |
Data Science Master Virtual Internship from Altair - Eduskill 🔗 | Algorithms on strings and graphs by University of California San Diego 🔗